

Self-Exciting Jump Dynamics in Bitcoin Variance

Do jump clusters predict when the variance risk premium compresses?

1,838,880 one-minute bars, Jan 2023 – Jun 2026 · Deribit DVOL options data · all code and derivations in repository

The question. Bitcoin's price jumps arrive in clusters, like earthquakes and their aftershocks. If a cluster is running, is the options market's variance premium about to compress? Answering that needs a latent quantity nobody observes — the *jump-variance state* — so the project builds the machinery to extract it, verifies that machinery works, and then tests the trading question honestly.

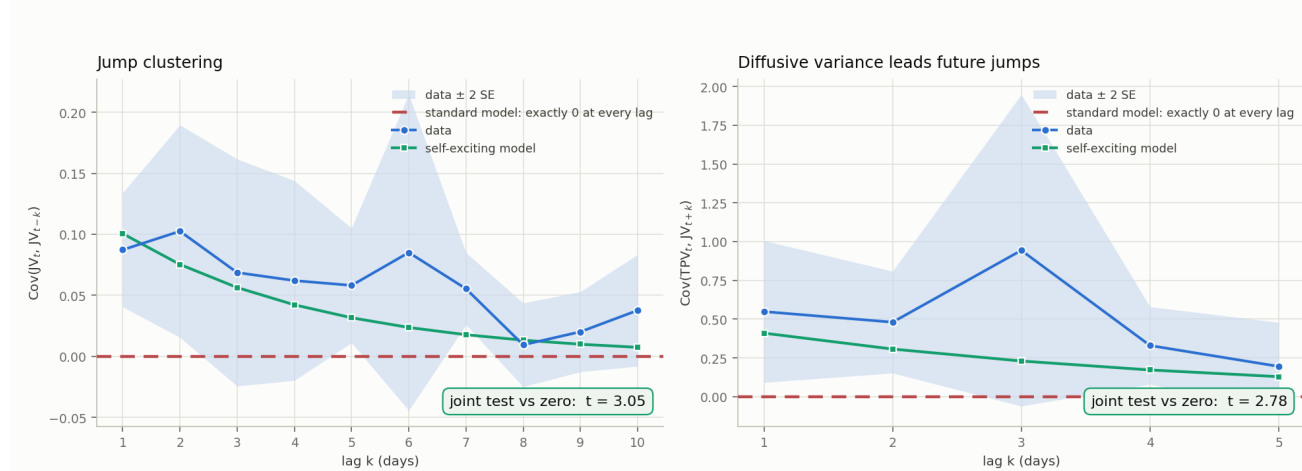
1. What I found

1. Bitcoin jumps cluster, and the standard model cannot accommodate it. The textbook constant-intensity jump-diffusion predicts that jump energy on disjoint days is independent: the JV autocovariance is *exactly zero at every lag, for every parameter value*. It is not.

model-free test	standard model predicts	data
Jump clustering, $\text{Cov}(\text{JV}_t, \text{JV}_{t-k})$, joint over 10 lags	0	$t = \mathbf{3.05}$
Diffusive variance leading future jumps, joint over 5 lags	0	$t = \mathbf{2.78}$
Lag-1 jump clustering alone	0	$t = \mathbf{3.74}$

Bitcoin jumps cluster — the standard model says they cannot

Constant-intensity jump-diffusion predicts both curves are identically zero for every parameter value. Both are significantly positive.



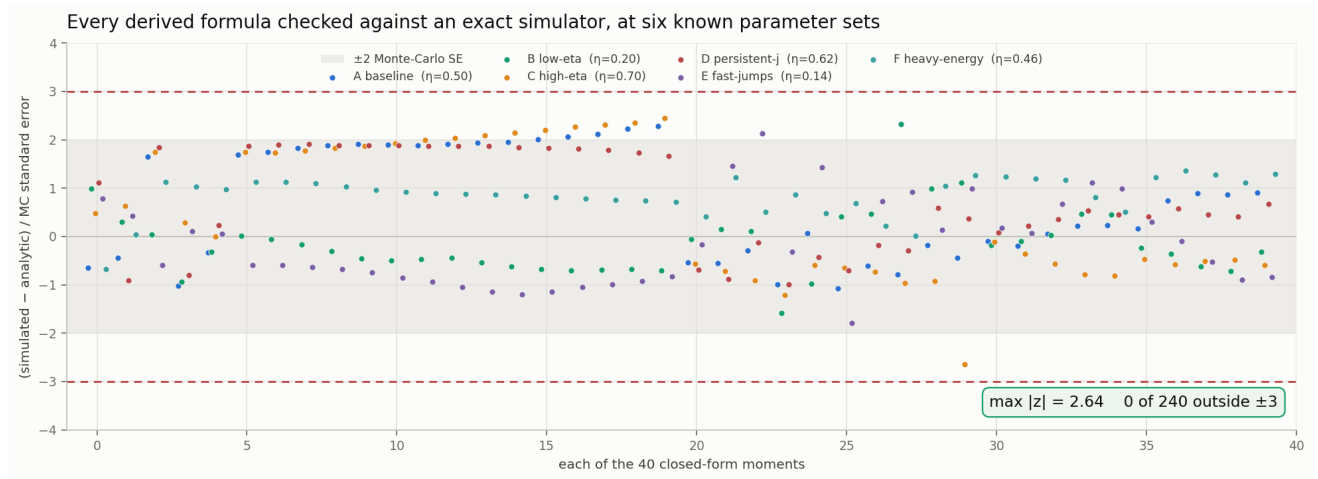
2. A self-exciting (Hawkes) repair fits. Letting the jump arrival rate depend on the jump-variance state, $\lambda(t) = \lambda_0 + \lambda_1 V^j(t^-)$, reproduces both curves. Calibrated by GMM on 40 moments: only **2 of 40** moments miss by more than two standard errors — what chance alone produces — and the misses are scattered rather than concentrated in any block. Point estimates: about **24%** of jumps are aftershocks of earlier jumps ($\hat{\eta} = 0.24$), individual shocks decay with a 1.8-day half-life while *clusters* persist 2.4 days.

3. Bitcoin variance carries a large risk premium. Option-implied variance exceeds the model's physical expectation by **20% on average** and on **65% of days**; it exceeds subsequently realised variance by **25%**. The premium mean-reverts with a **13.8-day half-life**.

4. The model still under-states persistence. On held-out crash events, variance stays elevated for **9.5 days** against the model’s predicted **2.4**. An independent over-identification test points the same way (a $6.3\times$ gap), so something slow is still missing — most likely a third variance factor. The Hawkes repair moves the right direction but not far enough.

2. How I know the machinery is right

Before touching real data, the estimator was proved on data whose answers were known. An exact simulator (Ogata thinning for self-exciting jumps, noncentral χ^2 for the CIR factor, analytic within-day integrals — no discretisation) generated $\geq 50,000$ days at six known parameter vectors.



Every analytic moment landed inside Monte-Carlo error ($\max |z| = 2.64$, none of 240 beyond ± 3); GMM recovered the true parameters; and 95% bootstrap intervals covered at 0.86–0.98 across 50 independent datasets.

This process caught four real bugs, each of which would have silently corrupted the results:

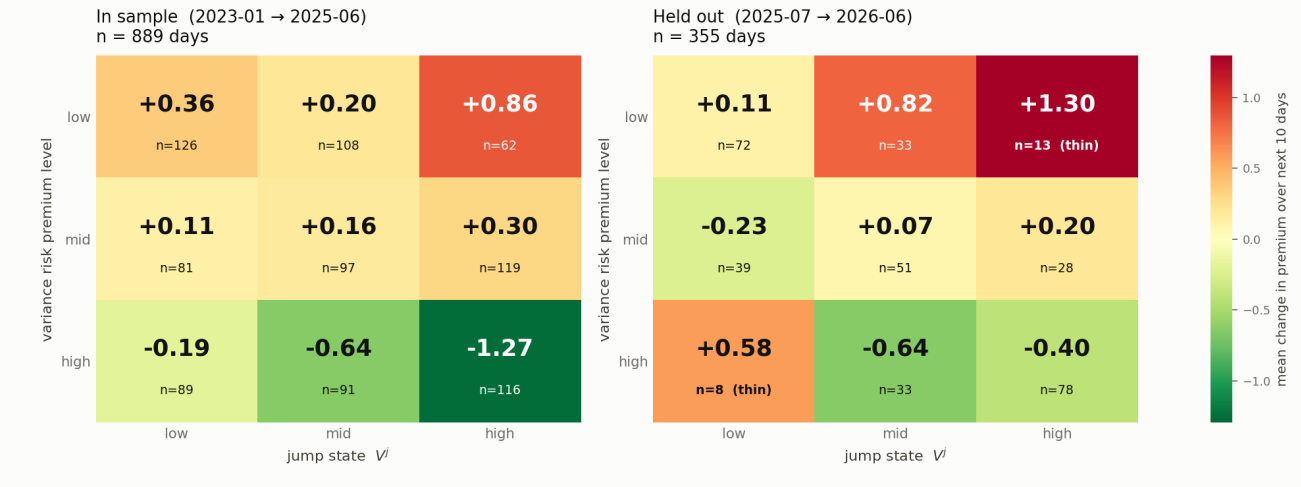
1. Tripower variation used exponent $4/3$ (quarticity) rather than $2/3$ (variance) — the “jump variance” series was really *total* variance. Caught by cross-checking against a second robust estimator (MedRV), which now agrees at $\rho = 0.996$.
2. The block bootstrap re-sliced the raw series and recomputed autocovariances, splicing unrelated days together. Interval coverage for κ_c was 0.00; resampling moment-contribution rows instead restored it to 0.93.
3. Shrinking the GMM weighting matrix toward a scaled identity ignored that moment variances span five orders of magnitude, starving the jump-clustering moments of weight. It had inverted the persistence estimate ($\kappa_j = 3.43$ versus the correct 0.29).
4. One of the two over-identifying “tests” was an algebraic tautology of the parametrisation — it holds for arbitrary parameters and tests nothing. It only has content when the coefficients are estimated freely from separate moment blocks, which is how it now runs.

3. The trading question, and the honest answer

Sorting every day by how rich the premium is *and* how hot the jump state is produces a large, ordered in-sample effect. Holding the premium rich, a hot jump state compresses it roughly $20\times$ harder than a quiet one over the following ten days, and raises the odds of a large compression by $2.1\times$.

A large in-sample effect that held-out data does not reproduce

The bottom row holds the premium rich, so any variation across it must come from the jump state.
In sample the row is ordered $(-0.19 \rightarrow -0.64 \rightarrow -1.27)$ — rich premium plus a hot jump state compresses hardest.
Out of sample the ordering breaks $(+0.58 \rightarrow -0.64 \rightarrow -0.40)$, and the only cell favouring the effect holds 8 observations.



It does not replicate out of sample. On one year of held-out data the ordering breaks, and the single cell that would support the effect contains eight observations. With roughly 35 independent ten-day windows, the holdout can neither confirm nor refute — but it certainly did not confirm.

Three further results were discarded once tested properly, each an artifact rather than a signal:

- The premium appeared to predict variance-selling returns at $t = 5.7$ — but signal and payoff share the implied-variance leg ($\rho = 0.97$), and the effect vanishes once that is controlled ($t = 0.99$).
- The continuous-variance state appeared to predict premium changes at $t = 2.5$ — but the model's forecast is built *from* that state, a purely mechanical channel ($t = 8.5$). Against pure market data the effect is $t = 0.10$.
- The structural forecast does not beat a three-term HAR regression (R^2 0.117 versus 0.113).

4. Method, briefly

Two latent factors drive total variance: a persistent Brownian-driven CIR component and a jump-driven component whose arrival intensity is affine in its own level, making jumps self-exciting (equivalent to a marked Hawkes process). Because the intensity is affine, all 40 unconditional moments — means, variances, autocovariances and two-sided cross-covariances of daily tripower variation and jump variance — remain in closed form, and are matched to their sample analogues by GMM with an analytic 40×9 Jacobian, Ledoit–Wolf regularised weighting, and moving-block bootstrap intervals. Parameters are estimated on 2023-01 to 2025-06 only; 2025-07 onward, containing the October 2025 crash, is never used in estimation.

5. What I would do next

The persistence shortfall (9.5 days observed against 2.4 predicted) and the $6.3\times$ over-identification gap point at the same omission, so the natural extension is a third, slower variance factor. Separately, the compression result needs more held-out data than one year to be settled; the sample, not the method, is the binding constraint.

Reproduce everything with `python run_all.py`. Full derivations in `docs/affine_intensity_theory.pdf`; estimator verification in `docs/estimation_verification_report.pdf`.